QF1100 Midterm Cheatsheet

1. THEORY OF INTEREST

Accumulation & Compounding

- **Accumulation Function** $a(t)$**:** Value of \$1 at time t. $a(0) = 1$. Value at $t = \text{Principal} \times a(t)$.
- **Simple Interest:** $a(t) = 1 + rt$. Linear growth. Earns interest only on principal.
- **Compound Interest:** $a(t) = (1 + r)^t$. Exponential growth. Earns interest on interest.
- **Nominal Rate** $r^{(p)}$**:** Annual rate compounded p times a year. Interest per period $= r^{(p)}/p$.
- **Effective Rate** r_e**:** Actual interest in a year. $1+r_e = (1+r^{(p)}/p)^p$. For $p > 1$, $r_e > r^{(p)}$. As p increases, r_e increases.
- **Equivalent Rates:** Yield same accumulation over 1 year. $(1 + r^{(p)}/p)^p = (1 + R^{(q)}/q)^q$.

Continuous Compounding & Force of Interest

- **Continuous Rate:** As $p \to \infty$, $a(t) = \lim_{x\to\infty} \left(1 + \frac{r}{x}\right)^x = (e^r)^t$.
- **Equivalent Rates:** Yield same accumulation over 1 year. $(1 + r^{(p)}/p)^p = (e^r)^t = 1 + r_e$.
- **Force of Interest** $\delta(t)$**:** Infinitesimal relative rate of change. $\delta(t) = \frac{a'(t)}{a(t)} = \frac{d}{dt}[\ln(a(t))]$.
- **Accumulation with** $\delta(t)$**:** $a(t) = \exp(\int_0^t \delta(u)du)$.
  $a(s,t) := \frac{a(t)}{a(s)} = \exp\left(\int_s^t \delta(u)du\right) = \frac{\exp(\int_0^t \delta(u)du)}{\exp(\int_0^s \delta(u)du)}$
- **Constant** δ**:** If $\delta(t) = \delta$ (constant), then $a(t) = e^{\delta t}$. Here, $\delta = \ln(1 + r_e)$.
- **Consistency Principle:** $a(t_0, t_n) = a(t_0, t_1)a(t_1, t_2) \dots a(t_{n-1}, t_n)$.

Present Value & Equivalence

- **Present Value (PV):** Value today ($t = 0$).
  $PV = \sum_{i=1}^n \frac{c_i}{a(t_i)} = \sum_{i=1}^n \frac{c_i}{(1+r)^{t_i}}$. Discounting brings future value to present.
- **Time Value (**TV_t**):** Value at time t. $TV_t(C) = PV(C) \times a(t) = TV_t(C) = \sum_{i=1}^n \frac{c_i}{(1+r_i)^{t_i}} \cdot (1 + r)^t = \sum_{i=1}^n \frac{c_i}{(1+r_i)^{t_i-t}}$.
  Also, $TV_t(C) = \frac{a(t)}{a(s)} \cdot TV_s(C) = TV_s(C) \cdot (1 + r)^{t-s}$.
- **Principle of Equivalence:** Two cash flows are equivalent if they have the exact same PV.
- **Net Present Value (NPV):** PV of all inflows and outflows. A measure of value created today. Invest if $NPV > 0$. A is superior to B if NPV(A) > NPV(B).

Internal Rate of Return & Equation of Value

- **Equation of Value:** Set $NPV = \sum_{i=0}^n \frac{c_i}{(1+r)^i} = 0$ for the cash flows.
- **Internal Rate of Return (IRR):** Any non-negative root r of the equation of value. It is independent of prevailing market rates.
- **Uniqueness Theorem:** IRR is UNIQUE IF: (1) Initial cash flow $c_0 < 0$ (outflow), (2) All subsequent $c_k \geq 0$ (inflows), and (3) Total sum of cash flows $\sum_{i=0}^n c_i > 0$.
- **Decision Rule:** Invest if IRR > required return rate. Use NPV in conjunction with IRR. A is superior to B if IRR(A) > IRR(B).

Newton-Raphson Method

- Used to estimate IRR or bond yield when $n > 2$. Requires function to be differentiable. Has quadratic rate of convergence.
- **Formula:** $\alpha_{n+1} = \alpha_n - \frac{f(\alpha_n)}{f'(\alpha_n)}$.
- **Steps:** (1) Write PV equation $f(x) = 0$ where $x = \frac{1}{1+r}$. (2) Find bounds a, b where $f(a)$ and $f(b)$ have opposite signs. (3) Guess x_0. (4) Iterate.

Annuities (Constant Payment A)

- **Ordinary (Arrears):** Payments at END of periods.
  $PV(OA) = \sum_{t=1}^n \frac{A}{(1+r)^t} = \frac{A}{r}[1 - \frac{1}{(1+r)^n}]$.
- **Due (Advance):** Payments at BEGINNING of periods.
  $PV(AD) = \sum_{t=1}^n \frac{A}{(1+r)^t} = \frac{A(1+r)}{r}[1 - (1 + r)^{-n}] = PV_{OA} \times (1 + r)$.
- **Deferred Annuity:** Calculate PV at start of annuity (e.g., $t = k$), then discount back to $t = 0$ by multiplying by $(1 + r)^{-k}$.
- **Perpetuity Ordinary / Immediate:** Infinite payments.
  $PV = \sum_{t=1}^\infty \frac{A}{(1+r)^t} = \frac{A}{r}$.
- **Perpetuity Due:** $PV = \sum_{t=0}^\infty \frac{A}{(1+r)^t} = \frac{A(1+r)}{r}$.
- **Varying Annuities:** If payments change, split into two separate annuity streams or geometric progressions.

Loans & Repayment

- Loan amount $L = PV(\text{installment payments}) = PV(C)$.
- $L_m^{Balance}$: immediately after the mth installment been paid is TV_m.
- **Prospective Balance:** PV(remaining payments) $t > m$.
  $L_m^{Balance} = \frac{A}{r}[1 - \frac{1}{(1+r)^{n-m}}]$. (Does not need L).
- **Retrospective Balance:** Accumulated loan minus accumulated payments.
  $L_m^{Balance} = L(1 + r)^m - \frac{A}{r}[(1 + r)^m - 1]$. (Requires L).
- **Drop Payment (Non-integer** n**):** Make $\lfloor n \rfloor$ full payments plus smaller final payment B. Find integer periods, subtract PV of full payments from L, then accrete remaining balance to the time of payment B.

2. BONDS & TERM STRUCTURE

Bond Basics & Risks

- P: Current Price(Listed on market), F: Face/Par Value(coupon based on this), R: Redemption Value.
- c: Nominal coupon rate, m: Periods/year.
- n: Total periods, λ: Nominal yield.
- **Coupon Payment:** cF/m.
- **Law of One Price:** Price = PV of all future cash flows.
- **Risks:** Default risk (failure to pay), Liquidity risk (cannot find buyer), Reinvestment risk (inability to invest coupons at required return).

Bond Pricing Formulas

- **General Price:** $P = \frac{R}{(1+\lambda/m)^n} + \sum_{i=1}^n \frac{cF/m}{(1+\lambda/m)^i}$.
- **If** $F = R$**:** $P = F\left[\frac{1}{(1+\lambda/m)^n} + \frac{c/m}{\lambda/m}\left(1 - \frac{1}{(1+\lambda/m)^n}\right)\right]$
  $= F + F\left(\frac{c-\lambda}{\lambda}\right)\left[1 - \frac{1}{(1+\lambda/m)^n}\right]$
  Typical that F = R (Unless otherwise stated)
- **Makeham Formula:** $P = K + \frac{c}{\lambda}(F - K)$, where $K = \frac{R}{(1+\lambda/m)^n}$ (PV of redemption value).
- **Recursive Price:** $P_{k+1} = P_k(1 + \lambda/m) - cF/m$ (Relates price between 2 successive periods).

Bond Types & Behaviors

- **Premium:** $P > F \iff c > \lambda$.
- **Par:** $P = F \iff c = \lambda$.
- **Discount:** $P < F \iff c < \lambda$.
- **Zero-Coupon:** $P = \frac{R}{(1+\lambda)^N}$ (No coupons).
- **Perpetual (Consol):** $P = \frac{cF}{\lambda}$ ($n \to \infty$) (never matures).
- **Singapore Savings Bonds (SSB):** Non-tradable, rates vary by year, early redemption lowers yield.
- **SGS Bonds:** Tradable on market, no early redemption (must sell).
- **Price-Yield Curve:** Price P is a decreasing, convex function of yield λ. Max price is at $\lambda = 0$ (sum of all undiscounted cash flows).

Macaulay Duration (D)

- Weighted average time to maturity. Measures price sensitivity to interest rate changes.
- **Weights:** $w_i = \frac{PV(c_i,t_i)}{PV(C)}$. Higher PV means more weight placed on that specific time t_i.
- **General:** $D = \frac{\sum t_i PV(c_i,t_i)}{PV(C)} = \sum w_i t_i$.
- **Bounds:** $t_1 \leq D \leq t_n$.
- **Zero-Coupon Bond:** $D = T$ (Maturity time).
- **Duration:** $D = \frac{1}{P}\left[\sum_{i=1}^n \left(\frac{(i/m)(cF/m)}{(1+\lambda/m)^i}\right) + \frac{(n/m)R}{(1+\lambda/m)^n}\right]$,
  where $P = \sum_{i=1}^n \left(\frac{cF/m}{(1+\lambda/m)^i}\right) + \frac{R}{(1+\lambda/m)^n}$.
- **Bond Redeemable at Par (**$R = F$**):**
  Let $\mu = \lambda/m$ and $\Upsilon = c/m$.
  $D = \frac{1+\mu}{m\mu} - \frac{1+\mu+n(\Upsilon-\mu)}{m\Upsilon[(1+\mu)^n-1]+m\mu}$.
- **Bond Priced at Par (**$P = F$**, so** $c = \lambda$**, so** $\mu = \Upsilon$**):**
  $D = \frac{1+\mu}{m\mu}[1 - \frac{1}{(1+\mu)^n}]$.
- **Perpetual Bond Duration:** $D = \frac{1+\mu}{m\mu} = \frac{1+\lambda/m}{\lambda}$, where λ is nominal annual yield.
- **Key Properties:** Duration is independent of the face value (F) and independent of the frequency of compounding.

Yield Curves & Term Structure

- **Term Structure:** Relaxes constant interest rate assumption. Rates vary by time-to-maturity.
- **Normal Yield Curve:** Upward sloping. Optimistic. Seen during economic expansions.
- **Inverted Yield Curve:** Downward sloping. Rare. Seen during or just before recessions/high inflation.
- **Spot Rate (**s_t**):** Annual interest rate for money held from $t = 0$ to t. $a(t) = (1 + s_t)^t$. It is the yield of a zero-coupon bond.
- **Forward Rate (**f_{t_1,t_2}**):** Interest rate observed at t_1 lasting until t_2.
- **Rate Relationship:** $(1 + s_k)^k = (1 + s_j)^j(1 + f_{j,k})^{k-j}$.
- **From yield of zero-coupon bond:** $P = \frac{R}{(1+s_n)^n} \to$
  $s_n = (\frac{R}{P})^{1/n} - 1$.

Bootstrap Method

- Used to estimate missing spot rates using available coupon-paying bonds.
- **Step 1:** Use 1-period zero-coupon bond to find s_1: $P_1 = \frac{R}{(1+s_1)}$.
- **Step 2:** Use 2-period coupon bond to find s_2: $P_2 = \frac{cF/m}{1+s_1} + \frac{cF/m+R}{(1+s_2)^2}$.
- **Step 3:** Repeat iteratively to find $s_3, s_4 ... s_n$.

Arithmetic and Geometric

- **Arithmetic Progression:** $S_n = \frac{n}{2}\left[2a + (n-1)d\right]$.
- **GP(n terms):** $S_n = a\frac{1-r^n}{1-r}$.
- **GP(infinite terms:)** $S_\infty = \frac{a}{1-r}$.